

Homework — 1.1.2 Types of Processor

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.1.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Describe **three** differences between **RISC** and **CISC** processors. **[3]** (AO1)

2. A smartphone manufacturer uses a **RISC (ARM)** processor rather than a **CISC (x86)** processor. Give **two** reasons, linked to a smartphone, why RISC is the better choice. **[2]** (AO2)

3. In a **CISC** processor the complexity sits in the **hardware**; in a **RISC** processor it sits in the **software/compiler**. Explain **one** consequence of this difference for **pipelining**. **[2]** (AO1)

4. A GPU is well suited to carrying out the **same operation on many items of data at the same time**. Explain, with reference to the **structure** of a GPU, why this is the case. **[3]** (AO1)

5. A video-editing application is rewritten to spread its work across **all the cores** of a multicore CPU, yet the speed-up is far less than the number of cores would suggest. Explain **two** reasons why the speed-up is limited. [2] (AO2)

6. GPUs are increasingly used for **general purpose computing** — tasks that have nothing to do with producing graphics. Give **two** examples of non-graphics tasks for which a GPU is well suited. [2] (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. Explain why the **Von Neumann** architecture can suffer from a performance **bottleneck**, and state how the **Harvard** architecture avoids it. [3] (retrieval from 1.1.1)

8. State **two** factors (other than number of cores) that affect the performance of a CPU, and for each give its effect. [2] (retrieval from 1.1.1)

9. Name the register that holds the **current instruction** while it is being decoded and executed. [1]
(retrieval from 1.1.1)
